PART 2 OF 11 · LAYER 1: KEEP THE HOUSE DRY

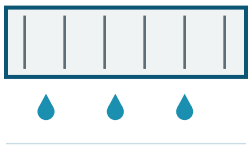
Humidity and dew point

THE SIMPLE VERSION

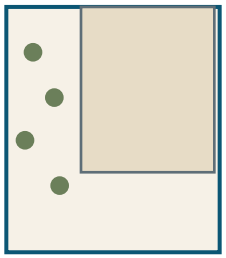
Water hides in the air. We cannot see it.
When wet air touches something cold, the water comes out as little drops.
We keep the air dry so the drops never show up.



Cold glass
drops appear on the outside



Cold AC vent
can drip when indoor air is too humid



Cool, hidden surfaces
stay damp longest

Dew point is the temperature where water vapor turns back into liquid. Any surface colder than the air's dew point will get wet.

Measure, then manage

- **Use a digital hygrometer** in the sensitive person's bedroom and one in the main living area. Check them morning and evening for the first few weeks so you learn your home's pattern.
- **Check that it reads true.** Seal the hygrometer in a zip bag with a cup of damp table salt for 8 to 12 hours. It should read about 75 percent; note how far off it is and correct future readings by that amount.[1]
- **Treat readings that stay above 55 percent as a problem to correct**, not just a comfort issue. ACAAI advises staying below 55 percent and using a dehumidifier in humid climates.[2]

Dew point: the number building scientists watch

Relative humidity changes with temperature, but **dew point** tells you how much water is actually in the air. In cooled buildings, keeping the indoor dew point below about 55°F nearly always keeps surface humidity below 80 percent, even on cool surfaces. In contrast, air at 78°F and 55 percent RH will push any surface cooler than about 66°F above 80 percent.[3] Many smart thermostats and hygrometers display dew point.

Room temperature	At 40% RH	At 50% RH	At 60% RH	What it means
72°F	46°F	52°F	57°F	Lower setpoints need drier air, or vents and nearby surfaces can sweat.
75°F	49°F	55°F	60°F	50% RH at 75°F lands right at the 55°F dew point target.
78°F	52°F	58°F	63°F	Warmer homes can hold a bit more RH; watch cool spots and closets.

Indoor dew points calculated with the Magnus formula. Values are rounded to the nearest degree.

Your next step

Place a hygrometer in the bedroom and the main living area, and write down the readings for one week. If humidity stays above 50 percent, see Part 3 (AC settings) or contact us.

Sources for this part

1. Canada Mortgage and Housing Corp., Measuring Humidity in Your Home.
https://publications.gc.ca/collections/collection_2011/schl-cmhc/nh18-24/NH18-24-1-2009-eng.pdf
2. ACAAI, Dust Allergies.
https://acaai.org/allergies/allergic-conditions/dust-allergies/
3. ASHRAE Handbook, Moisture Management in Buildings (Harriman and Lstiburek; ASHRAE Standard 160 criteria).
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